

1st class

02-02-2026

Math 3141

x

□ Statistical - is a science of collecting, organizing, presenting & analyzing, interpreting, Numerical & Non Numerical data of a problem to arrive at an effective decision

• Types of Statistical - Statistical is two type.

→ যার data নিয়ন্ত্রণ করে

① descriptive Statistics

② inferential Statistics.

সম্প্রদায়িক (sample নিয়ে কাজ করে)

→ descriptive

□ Population is the collection of all possible objects or individuals of interest.

Ex: All the student of Varendra University.

All the tiger of Sunderban.

→ inferential

□ Sample is a part or portion of population.

Ex: The student of CSE of VU.

Ten Tigers of Sundarban.

Variable থেকে data পাঠে।

www.ahammed.vu.wixsite.com
profile
বই, CO দেখা আছে দেখে নিবে

Variable এর definition, type দেখে আয়বে।

Variable - is a type of symbol used to represent a number that can change.

Statistic for all
এই book দেখবে for
Math

Types - 2 types

- Qualitative
- Quantitative

Q. Application of specific statistics in CSE

⇒ A variable is a symbol that represents an unknown or changing value in mathematics.

In CSE statistics is used everywhere. Applications:-

- Student result analysis.
- Website average response time.
- Machine Learning.
- Artificial Intelligence.

2nd class

Statistics এর Basic

HomeTask - কত ধরনের Statistics

Descriptive StatisticsChart আছে? কী কী
বাজে লাগে।

- Collect data
e.g - Survey
- present data
e.g - Tables and graphs.
- Characterize data
e.g - Sample mean = $\frac{\sum x_i}{n}$

Inferential Statistics ⇒

- Estimation - e.g Estimate the population mean weight using the sample mean weight.
- Hypothesis testing

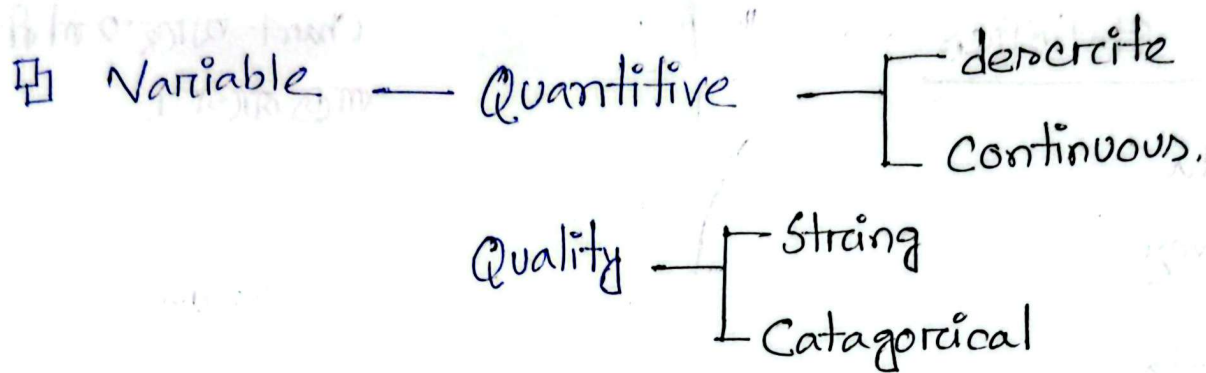
* Sample নিয়ে হোকা যায় Inferential টা! (Analysis করা হয়)

Parameter & Statistic → Sample এর উপর কাজ করে।

→ A parameter is a summary measure computed describing a characteristic of the population.

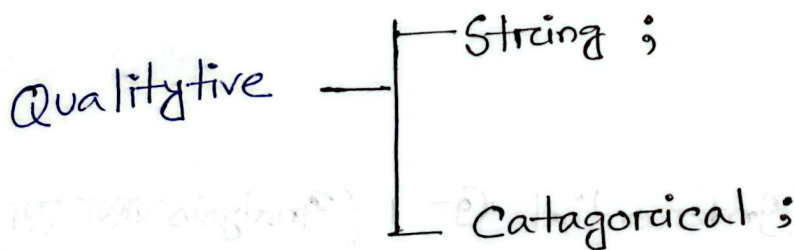
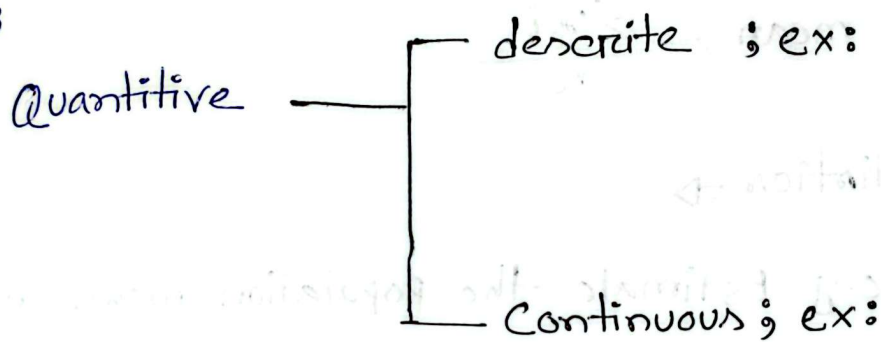
e.g - Average of NU student.

→ e.g - Average of the student of CSE dept of NU.



• Variable is a Characteristics or Condition that can Change or take on different values.

Example :



স্বাধীনতা & fraction অহ অব নিতে পারে যেগুলো Continuous Variable
" " " " নিতেপারে না । যেমনঃ বেক, চেয়ার, family
যেটা describe
এগুলো স্বাধীন

☞ There are four level of data :-

Nominal → Gender, Eye color, Religious affection,

Ordinal → During a taste. ex: test of 4 soft drinks.

Interval → ex: Temp. on the fahrenheit scale. There is no natural zero point

Ratio → Numerical (*প্রতিটি measurement এর 10 টা করে ex. নিখরো).

☐ Functions of Statistics (Slide)

Limitation of Statistics :-

↳ কোনো একজন ব্যক্তির individual data নিয়ে বস্তু সম্বন্ধে নয়।

* CSE তে কখনো কখনো use হবে?

Online-2

Chap - 2

Date: 09-02-2026

☐ Frequency distribution & Chart Building —
↳ গনন অঙ্কন

- ① problem বের করা then
- ② collecting raw data →
- ③ Data organized →

Steps of ^{Constructing} frequency distribution :-

Step 1 : Calculating Range, $R = H - L$; Where, H = The highest value
L = The lowest value.

Step 2 : Calculating no. class k such that,

$2^k > N$ (For minimum value of k) $\rightarrow$ যদি থাকে $N=306$ value

k = positive integer value.

তাহলে k এর মান হবে
 $2^5 = 32$

N = No. of observations.

OR, H.A Sturges formula,

$$k = 1 + 3.322 \log_{10} N$$

$\rightarrow$ পূর্ণ সংখ্যা করে নিতে হবে
Round করে নিতে হবে

Step 3 : Length (or), size of class (c)

$$\text{Interval, } c = \frac{R}{k}$$

(Round UP) $\rightarrow 4.1$ থাকলেও 5 করে
নেবে।

Ex : Dr. Tillman এর টা

problem : Construct a frequency distribution table using suitable Length of class Interval of the following data.

Weight (kg) - 90, 12, 80, 100, 85, 90, 95, 30, 35, 60, 89, 120, 70, 90, 45, 15, 18, 20, 25, 35, 40, 60, 75, 99, 110, 45, 90, 81, 42, 85, 15, 18, 14, 20, 25, 48, 50, 30, 20, 30.

Solⁿ : Total number of observation $N = 40$.

highest value, $H = 120$

Lowest value, $L = 12$

$$\text{Range, } R = H - L \\ = (120 - 12) = 108$$

No. of class, k such that

$$2^k = 2^6 = 64 > 40$$

Therefore, No. of class, $k = 6$.

$$\text{Length or Size of class Interval, } C = \frac{R}{k} = \frac{108}{6} = 18$$

*fraction observation আমত পারে তাই এটাৰ সাথে 1 যোগ কৰাৰ নিত্য পাৰি

$\left\{ \begin{array}{l} \text{fraction or (a Round up)} \\ \text{পূৰ্ণ সংখ্যা আকলি (অকল) } \end{array} \right\} = 18 + 1 = 19$

Class Interval CI	Tally Mark	frequency (f_i)
$(12+19) \rightarrow 12-301$		14
$301-4850$ (30119)		8
$48-66$ $50-69$		3
$66-84$ $69-88$		6
84-102 $88-107$		7
102-120 $107-126$		2
120-		

Note : Lower Limit Included. $N = \sum f_i = 40$

1 যোগ করবেই class সংখ্যা ঠিক আসবে 6 টা

Note:-

for int. num $C = (+1)$

$N = f_i$ হবে
অন্য কিছু আছে

Lower Limit Included মানে, 12 lower টা নিয়েছি পরে 31 টা ২য় class
এর upper ও আছে ২য় class এর lower এ আছে ভয়ানক ভয়ানক 31 কে
২য় class এই লব lower দিয়েছে।

যদি continuous
4 class interval

নিব

যদি discrete
শাকলে discrete

নিব 1

সাক

CI	f_i
10-15	5
15-20	6
20-25	4

Gap তাই continuous
class.

CI	f_i
10-15	4
16-20	6
21-25	5

Gap আছে তাই

discrete class

যদি data থাকে,
10.5 তাই কে বইছে
নিব তখন continuous
এ নেওয়া সম্ভব

10.10 থাকলে dec. ৫
নেওয়া হবে অর্থাৎ con.
এর একই টা মানেই পাতা
টাতে পারবে।

* Lower limit
Included /

* Upper limit exclus.

২ টা
৩২ টা নিয়ে নিম্নলিখিত
হবে 1

$$\# \text{ Midpoint} = \frac{\text{Lower} + \text{Upper}}{2}$$

Relative frequency = total দিয়ে ভাগ করতে হয় । এর summation অবশ্যই 1 হবে ।

Cumulative frequency $\rightarrow (F_i)$
 ↓
 ক্রমযোজিত গনসংখ্যা

CI	f_i	F_i
10-15	4	4 ← যোগ 4+5=9
15-20	5	9 ← যোগ 9+6=15
20-25	6	

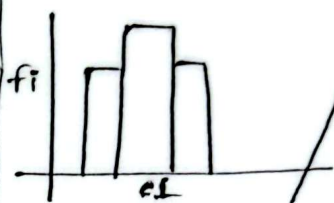
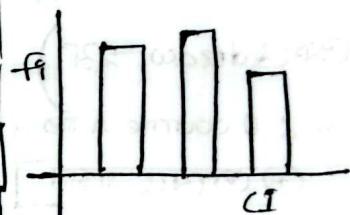
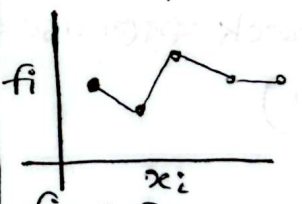
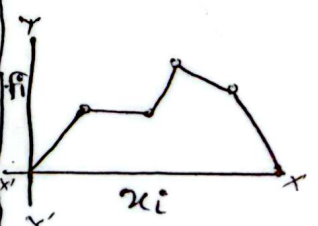
Relative frequency $\rightarrow (R_i)$

Midpoint $\rightarrow x_i$

Online

offline

Major Statistical Chart / Graph

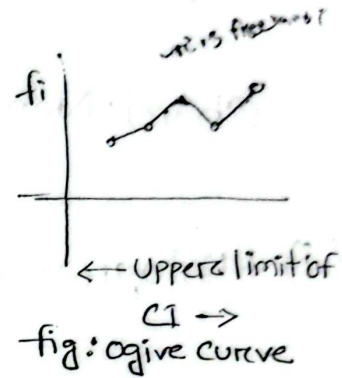
Chart Name	Data level in x axis	Data level in y axis	Structure
(i) Histogram	Class Interval (CI)	Frequency (f_i)	 fig: histogram
(ii) Bar-diagram	Class Interval (CI)	Frequency (f_i)	 fig: Bar-diagram
(iii) Frequency Curve	Midpoint (x_i)	Frequency (f_i)	for Discrete data → free hand  fig: frequency curve (con, Discrete data)
(iv) Frequency Polygon	Midpoint (x_i)	Frequency (f_i)	 fig: frequency polygon

(fig: 2.2) and (fig: 2.3) are drawn by free hand.

(V) Ogive Curve

Upper limit of CI

Cumulative frequency (f_i)



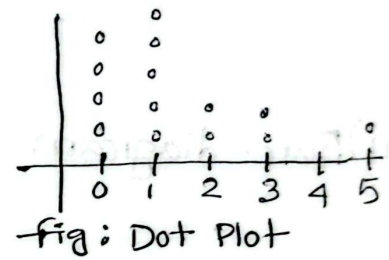
(VI) Dot Plot

(Discrete data এর ক্ষেত্রে draw হয়)

উদাহরণ: 0 course এর কতজন (+) পায়েছে স্বাক্ষর]

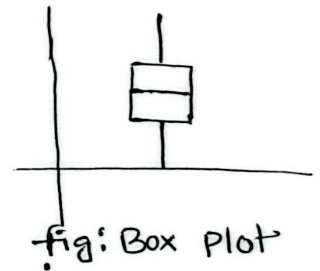
নেই

নেই

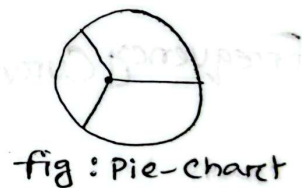


(VII) Box Plot

(Dataset এর outliers check করতে use করা হয়)



(VIII) Pie - Chart



H.W.

Write down the name of ^{Statistical} Chart with application &

Pattern/Chart.

(প্রায় 40 টার মধ্যে Chart আছে & কোম্পানি
অপ্লিকেশন হয় সেটা দেখাবে।)

⇒ Statistical chart

Data is two types ⇒ ① raw data

② Group data → গনন সহজ, fine, দেখা
হয় group data

Collection এর সময়-ভিত্তি করে data 2 types

① primary data ;

② Secondary data ; যু (যেকোনো data নিয়ে secondary, or BD bank থেকে data নেয়া
এমন type.

Date : 17.02.2026

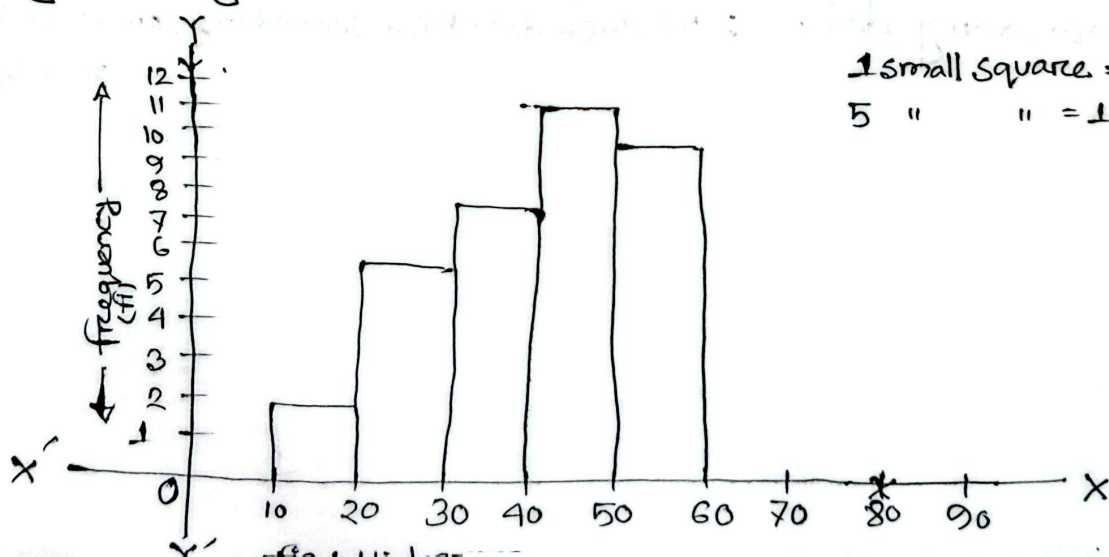
problem : Draw Histogram, frequency Curve, Frequency polygon and ogive curve of the following data.

Class Interval (CI)	10-20	20-30	30-40	40-50	50-60
frequency (f_i)	2	6	8	12	10

→ get graph data.

Solⁿ:

Let us consider XOX' be x -axis and YOY' be y -axis. we would like to include the class Interval (CI) in x -axis and the respective frequency in y -axis and then drawing histogram on a graph paper as follows -



1 small square = 1 unit in x -axis

5 " " = 1 unit in y -axis

Normal / Symmetric



this is ^{or Symmetric} normal graph

Non-Normal / Non-Symmetric

This graph is Non-Symmetric graph.

Frequency Curve: Let us consider XX' be x-axis and YY' be y-axis

Necessary Calculating Table

Class Interval (CI)	Midpoint (x_i)	frequency (f_i)	Cumulative frequency (F_i)
10-20	15	2	2
20-30	25	6	8
30-40	35	8	16
40-50	45	12	28
50-60	55	10	38

$$N = \sum_{i=1}^n f_i \\ = 38$$

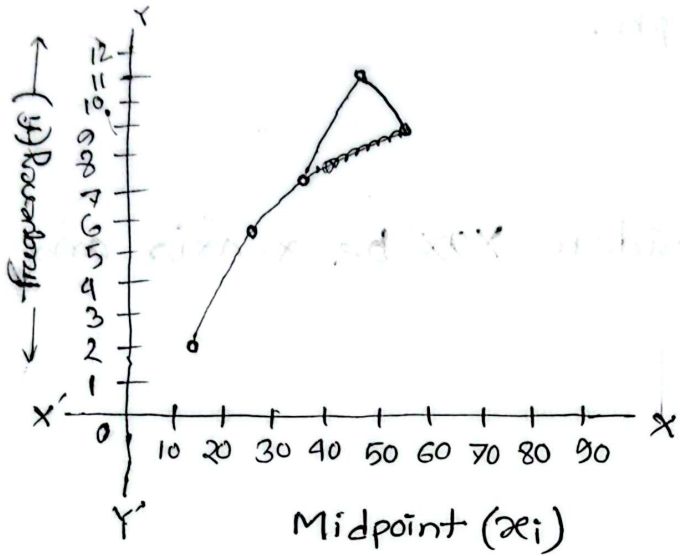
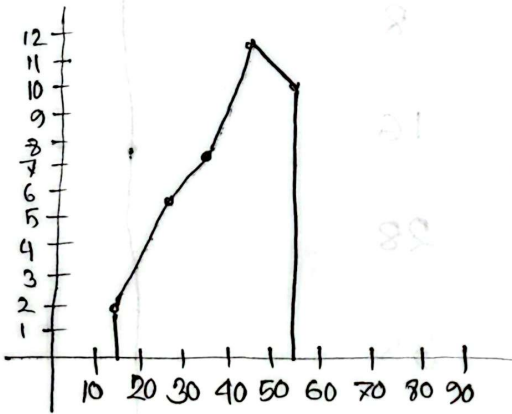


fig: Frequency Curve

at-free hand যোগ করবে

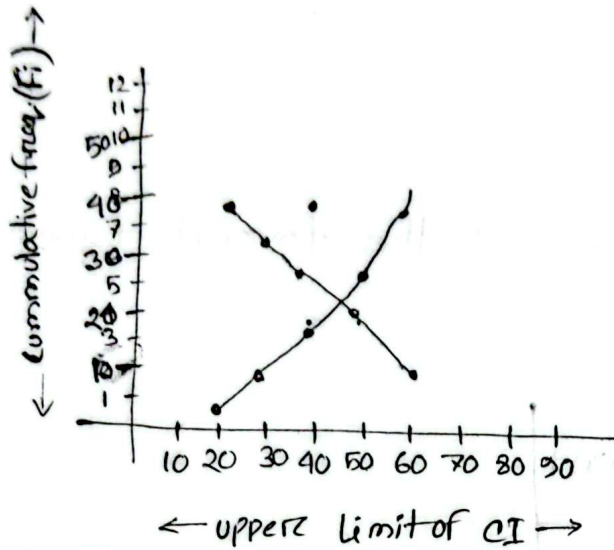
1 small unit square = 1 unit in x-axis
5 " " = 1 unit in y-axis

frequency Polygon : অব গুলোতেই Let us Consider -- নিখাত হবে ।
অবর ক্ষেত্র দিয়ে যোগ করতে হবে start & End point এ ঝিনিয়ে দিতে হবে or verticle
line টেন দিতে হবে



1 small square = 1 unit in x axis
5 " " = 1 unit in y-axis

Ogive Curve: - free hand & স্মিথিয়ান দ্বারা



1 small square = 1 unit in
x-axis

1 Small Square = 1 unit in
Y-axis

Class Interval	Upper limit	f_i	F_i
CI			
10-20	20	2	
20-30	30	6	
30-40	40	8	
40-50	50	12	
50-60	60	10	

CI	Midpoint	f_i	Cumulative freq.	
			Upper C.F	Lower C.F
10-20	15	2	2	38
20-30	25	6	8	36
30-40	35	8	16	30
40-50	45	12	28	22 22+8
50-60	55	10	38	10 22+10

Upper চার্টে লক্ষ্য
করতে হবে graph এ

(f_i) এর মাধ্যমে
নিচ থেকে যোগ
করবে

problem : The following data represent the favorite types of soft drinks as follows -

Types of Soft drinks class	No. of respondent
Pepsi	60
Co ca Cola	80
Mirinda	50
Mountain Dew	60
Fanta	30
Others	20

Draw Pie-Chart.

$$N = \sum f_i$$

$$= 300$$

Solⁿ: Necessary Calculating Table :-

Types of soft drink	No. of respondent	Relative frequency $R_i = \frac{f_i}{N}$	Amount in 360° $\theta_i = \frac{f_i}{N} \times 360^\circ$
Pepsi	60	$\frac{60}{300} =$	72°
co-ca Cola	80	$\frac{80}{300} =$	96°
Mirinda	50	$\frac{50}{300} =$	60°
Mountain	60	$\frac{60}{300} =$	72°
Fanta	30	$\frac{30}{300} =$	36°
Others	20	$\frac{20}{300} =$	24°
		$\sum R_i = 1$	$\sum \theta_i = 360^\circ$

Anti Clock wise করবে : চাই কি যে কবচের degree বুঝে থাকবে ।

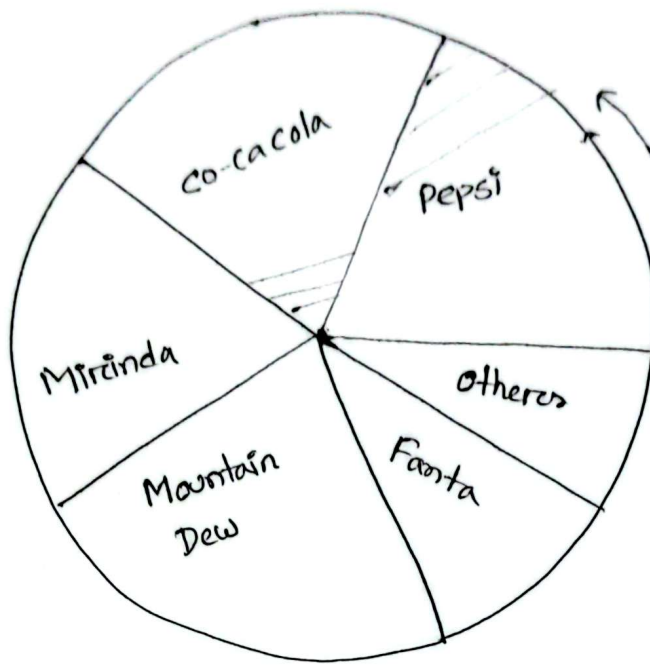


fig:- pie-chart-

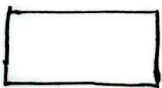
Legend $\Rightarrow$

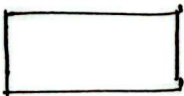
Pepsi - 

Co-cacola - 

Mirinda - 

Mountain dew - 

Fanta - 

Others - 

প্রকৃতি বা নিখিলেও হবে pie chart
এ-নিখে দিনই হবে ।